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REVIEW

STEP-NC AP238 - an excellent paradigm for smart manufacturing

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Abstract
Information exchange plays a crucial role in digital manufacturing. Newer paradigm such as Industry 4.0 utilizes data transfer through sophisticated technologies of sensors, internet, intelligent system, IoT and other related technologies across enterprise. Above concepts tends to create manufacturing sector into "Smart Industries". Exchange of product model information without human intervention helps in automation and integration of different sections of the manufacturing organization. STEP (STandard for Exchange of Product model data) used EXPRESS language, represents information of product model data in object oriented structure. STEP contains different Application Protocols (APs), which are suitable for different domains of product life cycle. AP203 represents geometrical and topological information of product. AP224 represents feature information of product in a well-defined structure. STEP-NC codes have data of manufacturing features as well as manufacturing process; it evolved as a superior format for computer control of machine tools as compared to G-M codes. STEP-NC was standardised in two forms, namely, ISO 14,649 and ISO 10303-238 (AP238). AP238 has rich set of information, which can be used across product lifecycle, as it uses STEP generic resources for building a particular model. STEP based methods and their application areas, seem to be an excellent conveyor of smart manufacturing or Industry 4.0, which would help in complete integration of CAD/CAPP/CAM. In this article, a detailed review of works related different STEP methods are presented along with detailed study of past research using different STEP formats. From this review study, it can be concluded that STEP AP238 provides an effective, efficient and sustainable format for seamless integration of various activities related to design and manufacturing processes involved in scenario of Industry 4.0 or Smart Manufacturing.

Keywords STEP-NC · AP238 · STEP based manufacturing · Industry 4.0

1 Introduction

Communication or information exchange becomes a crucial part in digital manufacturing. Today, manufacturing has been transformed from mechanical work to mechanization, then to automation and computer based automation. Now, it is digital manufacturing in which system become autonomous and governs machines through sensors, internet, Internet of Things (IoT) and servers. Cloud manufacturing is becoming a major part of various industrial sectors. Industrial Internet of Things (IIoT) utilizes dedicated sensors for specific applications in manufacturing and its process. Manufacturing equipment comprises of CNC machines, industrial robots, automated guided vehicles (AGVs), automatic material transferring, and handling, storing and packaging machines. These may be linked to servers via sensors through internet services and their data or operation progress can be seen in the integrated system of enterprise resource planning (ERP) software across the firm. Data exchange via computers plays a crucial role in the domain of Industry 4.0 or Smart Manufacturing.

With the advent of computers, processes in manufacturing industries become easier and faster. Earlier, on recommendations of design engineer, draughtsman draw design of the part to be manufactured, called as blueprints. On the basis of blueprints, manufacturing engineer prepares the

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required process plan to produce the finished part. Presently, Computer Aided Design (CAD) software helps not only for making drawing of the product either in 2D or 3D, but also assists for design analysis and for assembly simulation of the model. Computer Aided Engineering (CAE) gives opportunity to test the designed part within the specified conditions. Computer Aided Process Planning (CAPP) bridges the gap between CAD/CAM. Manufacturing modules of different CAD/CAM software can generate process plans and automatic numerical control codes can also be developed and verified through simulation interface of respective software.

The issues with CAD/CAM files importability and incompatibility were resolved with the most efficient neutral file format of STEP (Standard for Exchange of Product Model Data), which is more reliable format than IGES, SET, and PDDI [1]. Basically, STEP file provides representation of modelled or designed part in term of their geometry representations i.e. B-rep, CSG, etc., which is human readable and computer interpretable. Geometrical features in these file can be used to recognize manufacturing feature, which can transform design intent to manufacturing intent of particular product. Using these representation formats, manufacturing features can be identified such as hole, slot, pocket, step, etc., with the help of suitable data extraction and feature recognition algorithms. Features can be represented through two kinds of approaches, one is design-by-feature and other is feature recognition. Design-by-feature approach utilizes predetermined primitives elements and uses Boolean operators to describe a particular feature. Whereas feature recognition approach, utilizes geometry and topological data of CAD part along with suitable recognition technique to recognize or identify the features. Some of the important feature recognition techniques are syntactic pattern approach, logic based approach, and graph based approach, expert system approach and volume decomposition approach. The knowledge of manufacturing feature in a specified description would provide better understanding of product data model to the manufacturing engineer. With the knowledge of manufacturing features on modelled part an efficient, optimized and error free process planning shall be generated using CAPP system.

Extracted feature information should be mapped to different process planning activities. Feature based process planning involves selection of desired manufacturing operation, machine tools, process parameters, operation sequencing, machining conditions, tolerances etc., for a particular feature. CAPP uses manufacturing database for the selection of above production parameters and production rules. Automated NC codes shall be generated from the process plan of the part for machining, which can be used by CNC machines. STEP based methods and its application areas, seem to be an excellent conveyor of feature based process planning activities, which would help in complete integration of CAD/CAPP/CAM. G-M codes, the existing NC codes can be replaced using STEP-NC [2]. There were two formats in which STEP-NC was developed, first one is ISO 14,649 and another one is ISO 10303-238 [3]. Some of drawbacks of numerical control, G-M codes are as follows:

i. G-M codes are low level codes; they only provide tool cutter location data and do not provide the higher level information of manufacturing features and operations.
ii. G-M codes do not contain information of product geometry.
iii. G-M codes require post processor for translation of NC codes to tool/worktable positions.
iv. By looking at some lines of G-M codes, it is not possible to get information of type of operation/tools being used.
v. Some G-M codes are vendor specific NC codes and their nomenclature may be different on different CNC controllers.

Manufacturing industries are continuously improving their methods and techniques to produce low cost and high quality products. Faster processing of product data can enhance the effectiveness of industries for error free information interchange. Manufacturing information from the product data or designer intent should be effectively transformed. This can be successfully done through concept of feature via feature based process planning, which forms a vital link between CAD/CAM. Industries are nowadays focusing on the complete integration of their various departments related to CAD/CAPP/CAM. Numerical control codes i.e. G-M code lacks to represent efficient structure of high level information of feature and manufacturing operations. The broad domain of STEP standard using the STEP generic resources made it possible to cover entire product life cycle (i.e. design, manufacturing, utilization and disposal) through different Application Protocols. Feature recognition techniques from B-Rep of STEP AP203 or AP242 [4] file has given opportunity to various researchers to interpret manufacturing feature from CAD model. STEP AP224 [5] represents STEP feature information which is used to link feature based CAPP system. G-M code, which became an obstacle in intelligent manufacturing, can be replaced using STEP-NC. STEP-NC ISO 14,649 is based on Application Reference Model (ARM). STEP-NC AP238 [6] is based on Application Interpreted Model (AIM), which is formed using STEP generic resources.

STEP product data model is based on EXPRESS modelling language. The object oriented structure of EXPRESS provides a format in which information/data can be interpreted by computer. EXPRESS based structure of different

schemas of STEP Application Protocols provide an opportunity for excellent exchange of data related to CAD, CAPP and CAM linked entities. Therefore, STEP act as a very good conveyor of product data exchange throughout the lifecycle of product. Product data exchange involves an appropriate level of cyber security or privacy when exchanges through servers. Nowadays, industries are focusing on fully automated system, which can be achieved using concept of artificial intelligence and machine learning. These concepts are used in building a new manufacturing scenario of Industry 4.0 or the Smart Manufacturing (transformation from manual work to Industry 4.0 is shown in Fig. 1). It utilizes various domains of data exchange through servers, involving IoT devices using sensors across manufacturing firm. Industry 4.0, IoT, servers and cyber security are emerging technologies in manufacturing industries and are utilizing STEP based information exchange. Microcontrollers and microprocessor used in industrial machines are getting instructions from object oriented programming language based Integrated Development Environment (IDE), in which STEP format can provide seamless integration and interfacing across various components of above emerging technologies. Overall, the concept of Computer Integrated Enterprise and Distributed Manufacturing can be fully implemented using scenario of Industry 4.0, which would create a smart manufacturing system.

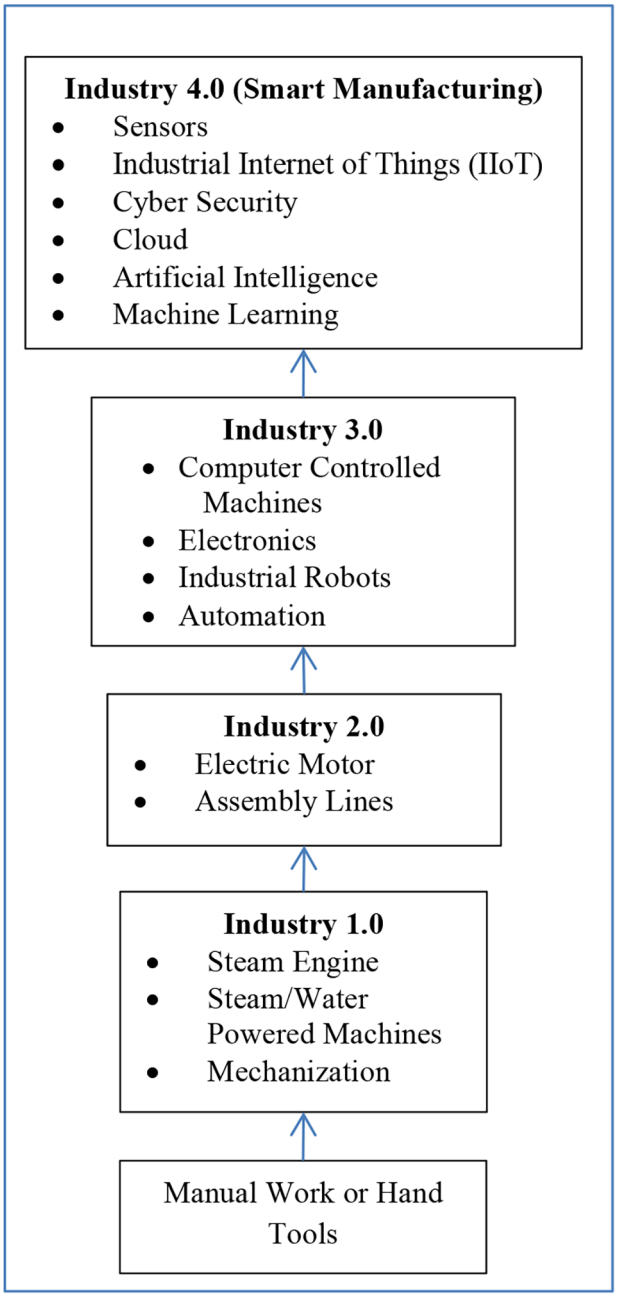


Fig. 1 Scenario from Hand Tools to Industry 4.0 (Smart Manufacturing)

In this article, an overall review of STEP based method is presented. This review would search for the functional capability of STEPs and its different format across various domain of manufacturing system and would also suggest a suitable STEP format to be used in the domain of Smart Manufacturing. A brief review in the field of feature recognition using STEP is presented here. However, exhaustive review of STEP and STEP-NC related to downstream activities of manufacturing is presented in this article.

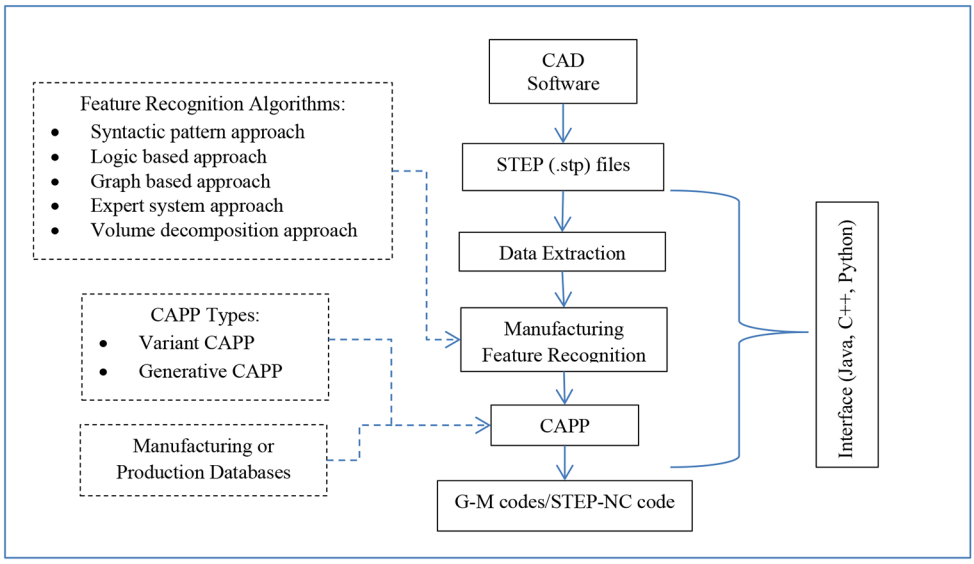
2 Work related to feature recognition and process planning using STEP (.Stp) format

STEP 203 file or AP242 of any design of solid model consists of geometrical features information such as points, line, vertices, edges, faces, solid using its B-rep or CSG details. Manufacturing feature can be recognized from these data models and can be also converted into STEP manufacturing information available in STEP AP224 and used to generate manufacturing downstream activities such as process planning and numerical control codes. CAPP is used to integrate CAD information to CAM information of product data model. Figure 2 shows the CAD/CAM integration using feature recognition algorithm and CAPP types to generate either G-M code or STEP-NC code using the interface to be developed using object oriented languages. Some of the important articles related to feature recognition and CAPP are mentioned in this session.

2.1 Feature recognition related review

Graph based technique proposed by Joshi and Chang [7] had given opportunity to researchers to use approach of Attributed Adjacency Graph (AAG) for recognizing features. The AAG concept of feature recognition consists of Attributed Adjacency Matrices (AAM) formed during evaluation of

Fig. 2 CAD/CAM Integration



concave and convex faces from the B-Rep data of solid model. For identification of features hint based approach along with AAG was used by McCormack and Ibrahim [8]. They considered interacting features as a single feature to search in AAG. From the interacting features, process planning was done to get the type of machining operation, tool and tool access direction for each feature. Sunil et al. [9] proposed a hybrid approach of feature recognition using graph (radial and serial structured) and along with ruled based approach from STEP based B-Rep CAD models. The hybrid approach avoids matching of sub-graph in feature recognition and also represents logical parent child relationships between features to automate downstream process planning activities. Bhandarkar and Nagi [10] did feature recognition from AP202 file (former version of AP203) of product model and used it to generate AP224. In their work, four types of form features (void, internal, external and connectivity modifying form features) were defined. Information from AP202 file, using ST-Developer™, along with sequential approach of B-Rep of features was used for feature recognition and represented the recognized features as AP224 entities. Zhang et al. [11] utilized hybrid algorithms of artificial bee colony and back propagation of neural network to recognize manufacturing features artificially. B. Venu et al. [12] recognized B-spline feature and its different sub-types for STEP file format. Complexity in linking AAG among various faces of model was simplified by Rameshbabu and Shunmugam [13]. In their article, volume subtraction approach along with face adjacency graph was used for feature recognition. The number of faces in the feature volume was reduced to simplify feature graph, so that wide variety of features could be identified through grouping of similar topological features. Further, setup planning using different rules for setup sequencing for various feature clusters was also reported through the rating and ranking of setup sequences.

2.2 CAPP related review

Different process planning activities for milling and drilling operation were generated using Feature-Based Inspection and Control System (FBICS) developed by Kramer et al. [14], which takes STEP AP224 ARM based file as input to the system. FBICS was implemented using the ALPS (A Language for Process Specification) process planning language to demonstrate feature based manufacturing. An AP224 implemented CAPP system based on Java was developed by You and Lin [15] for CAD/CAM integration. The object oriented structure of STEP feature model data was used to evaluate manufacturing process details stored in databases by servlet application over the network using the Enterprise Java Beans (EJBs). Kang [16] proposed a scheme to represent data in STEP AP224 based machining features, which benefits downstream users like process planner and inspection planner for making proper selection of process, processing conditions, etc.

Manafi et al. [17] proposed integrated manufacturing system to automate design information to manufacturing details which includes identification of tool approach direction, reference faces and represent dimensional and geometric tolerances of the machining features. For analysis of the early design stages and manufacturing evaluation Sharma and Gao [18] presented a product data management system, which was based on process planning and expert system. Miao et al. [19] automated CAPP system by integrating it with an I-DEAS (a commercial CAD/CAM system), which reduces time and user effort. Machining knowledge was used to extract machining features automatically from CAD

model in the CAPP system. This was used to eliminate manual selection of resources and process parameters on commercial CAM systems and hence automate the process. Gonzalez and Rosado [20] proposed a model with generality and flexibility in CAPP systems, which represent extraction of low-level information to structured knowledge in CAPP. They performed recognition of machining features without using geometrical entities such as machining volumes, quality requirements, etc.

The high level information of features such as tolerances, material specification, etc., can be effectively represented as STEP XML files compared to other CAD models. For the support of decision at early design stages Medani and Ratchev [21] proposed a new agent-based manufacturability environment based on STEP AP224. The cooperating agents (design, manufacturing, and facility planning agents) exchanged their structured message (exchange of information and requests) in XML format. Design by feature approach was used by Amaitik [22] for representation of AP224 feature information in XML format, which was given as input to an intelligent CAPP system to produce their corresponding machining operations. The inference engine of the intelligent CAPP system was implemented using hybrid approach of neural network and expert system. End milling and micro drilling process parameters can be obtained using extractor developed by Kumar et al. [23], which is based on STEP XML format and used approach of knowledge based and interactive feature recognition for micro parts of prismatic nature. Sheen and You [24] sliced the STEP based B-Rep model on the basis top and bottom surfaces and recognized Machining Features (MF), with different orientations. Using the MF information, manufacturing sequences was developed with different sets of tool paths. Houshmand and Valilai [25] discussed layered and modular platform for CAx collaboration named as Distributed-LAYMOD based on STEP standard. Their proposed platform used STEP modular architecture for distributed product data exchange via XML format.

Automatic generation of CAPP of prismatic part for manufacturing operation of rough and finished parts were developed using Machinable Volume Identifier (MVI) by Arivazhagan et al. [26], which shall be used as intermediate link for automated manufacturing. Their MVI utilized syntactic pattern approach for determination of manufacturing volume of features stored in volume databases. Creation of G-M codes through extraction of features on cylindrical parts were developed by Sivakumar and Dhanalakshmi [27] and they have also performed test part inspection. STEP-based platform was developed using the machining features accessibility and removable volumes by Nagarajan and Reddy [28]. They described features in the following three ways: single axis accessible features (e.g. hole and pocket), multiple axes accessible features (e.g. step and slot) and finishing features (e.g. chamfer, fillet and surfaces). Different feature relationships along with a set of rules were used for the determination of removable volumes and access direction. Different tool path and operation were generated using UniGraphics by Hou and Faddis [29] through their developed interface of Feature-based Machining Advisor (FBMach). Laguionie et al. [30] proposed Extended Manufacturing Integrated System (XMIS), to represent modularity and multi-directional high-level exchange of data between the modules for industrial equipment using STEP and STEP-NC data to reduce information loss. Their system helps in improving flexibilities and production setting times at the shop floor level.

3 Work related to CNC system and manufacturing using STEP-NC (ISO 14,649)

Shin et al. [31] used rule-based approach to represent the system functional architecture and developed algorithms for converting G-code into STEP-NC. Algorithm accuracy was tested using the developed prototype known as G2STEP system, which helped in reduction of time and cost through automatic conversion to STEP-NC from G-code. Reincarnation of planning level information was done through minimal requirement condition. The CNC was operated through PC using software interfaced with hardware and interface board. Suh et al. [32] developed architecture of shop floor programming known as PosSFP by implementing interpretation of STEP files for recognition of features, creation of process plans and generate and verify STEP-NC codes. Zhang et al. [33] proposed an autonomous STEP-NC controller, which consists of machining agent and planning agent for online planning at a level of shop-floor planning and the real-time planning with the help of feature based interface. Dharmawardhana et al. [34] developed CNC controller based on STEP-NC using Raspberry Pi and Arduino and canonical machining command was generated through interpreter. Xiao et al. [35] described a novel representation method for various resources to be used in a robot manufacturing system known as STEP-compliant Industrial Robot Data Model (IRDM). Their system helped in exchanging data between CAx systems and off-line programming system of robots for its kinematic and dynamic behaviours.

Lan et al. [36] presented a conceptual framework for STEP-NC controller interference with holonic compliant data to represent an intelligent CNC machine tool. The developed architecture comprised of different functional agents such as decision-making agent, process planning agent, interpreter agent, NCK/PLC agent, etc. The STEP-NC controller capabilities and its various function

of intelligent controller were fulfilled by these agents that co-operate within themselves in a desired manner. Suh and Cheon [37] represented architecture of an intelligent CNC system consists of three modules i.e. SFP/TPG (Shop floor programming/tool path generation) module, control module and common database module. They incorporated STEP-NC (ISO-14,649) data model for non-linear process planning to utilize the concept of STEP-compliant CNC. Lee et al. [38] developed XML based STEP-NC milling machine which includes STEP-NC interpreter. It consists of an intelligent controller which was used to develop tool path using the geometric and machining information from the STEP-NC data files.

Wang et al. [39] developed a system in which input was given in STEP-NC format and the system was interfaced with a CNC machine tool. For the CNC controllers, their system acted as a 'front-end' giving machine tools a 'Plug-and-Play' module. Their module consists of CNC native database to grasp the particular information of different CNC machines, which was used for mapping of the STEP-NC data to particular G-M code data. Lee and Bang [40] developed a tool path generator and motion controller for STEP-NC code in XML format. Motion control board and personal computer was used to develop controller for STEP-NC milling system. STEP Data Access Interface was used to develop Integrated Platform for Process Planning and Control (IP3AC) by Nassehi et al. [41] by Java programming language. In which they have develop 3D part and tree viewer and STEP-NC code generated were tested on SHOPMill machine of Siemens using interpreter for STEP-NC. Manufacturing process efficiency can be improved by deciding different equipment used in manufacturing activity during process planning at early stages and developing monitoring information from process plan itself for turning operation in STEP-NC, Shin et al. [42] developed STEP2M system (STEP to MTconnect) for virtual machining model. Agent based system (AB-CAM) based on STEP was developed by Newman et al. [43] for selecting different machining parameters from different databases from feature library and resource models and used to generate either STEP-NC code or G-M codes. Design by feature methodology was used to create feature in part model and developed the STEP compliant numeric code.

Due lesser information of data model of tool in ARM format of STEP-NC, STEP based tool data model (STEP-NCMtDm) was developed by Yang and Xu [44]. Their model was based on different data related to machining tool components. Performance, cost and kinematics of machining tool components was represented in EXPRESS language. In STEP-NCMtDm, a new schema, called machine_tool_information schema, was created and used along with the schema of ISO 14,649 to generate STEP-NC file in text format (Part 21). The generated STEP-NC file was tested on EMCO Concept Milling machine. For web based applications, XML format (Part 28) of STEP-NC was also developed. For conversion of NC codes (G-M codes) and STEP-NC codes to the CNC machine through its internal structure of Open Architecture Control (OAC), a module was developed by Yusof and Latif [45] using LabVIEW. Nassehi et al. [46] designed an object-oriented process planning system and utilized the concept called collaborative multi-agent systems of distributed artificial intelligence in a STEP-NC compliant environment for prismatic parts. The developed system was named as Multi-Agent System for Computer Aided Process Planning (MASCAPP), in which intelligent agents were used to represent manufacturing features information from STEP-NC codes and generates G-M codes. To reduce production cost, time and data loss, Zhao et al. [47] proposed a framework for STEP-NC enabled closed loop machining process to integrate CAD, process planning, machining and inspection chain. In their framework, the geometry and features data of STEP-NC were used for process-integrated inspection tasks, which focused on optimization of machining operations and quality of part. Garrido et al. [48] STEP-NC-based machine automation has been using the prototype implemented for saw processing of stones.

Brecher et al. [49] used STEP-NC based process chain for the integration of measuring technology that help in preservation of manufacturing process data and it also give feedback for the different activities of planning process. They presented prototype for closed-loop process chain scenario, which includes generation and execution of a STEP-NC program with a feedback to the CAM system for inspection tasks. For data sharing, manufacturing management, STEP-NC feedback and ERP, Danjou et al. [50] used OntoSTEP-NC system for demonstrating closed loop manufacturing. They used pocket feature of Example 1 of ISO 14,649 to provide CAM programmer different choices of process plan for interlinking CAM-CNC. STEP-NC code generation from G-M codes can be done using Universal Process Comprehensive Interface (UPCi) developed by Zhang et al. [51]. Bonnard et al. [52] used the STEP-NC data model for additive manufacturing and utilized the hybrid manufacturing approach of machining.

For milling and turning operations, Xu et al. [53] developed STEP-NC code from G-M codes using canonical machining functions. Nassehi et al. [54] discussed the application of mobile agents on STEP-NC based manufacturing chain. Mobile agents and multi-agents system were used to implement artificial intelligence in a manufacturing network which acts as enablers of interoperability in a global manufacturing enterprise. Data extraction from G-M codes (numerical control codes) were done by Zhang et

al. [55] using their developed algorithm for hybrid feature recognition. In that, UPCi was used to recognize feature and machining operation from generated STEP-NC codes and comparison for actual machining was performed on Siemen and Fanuc controlled machine. Kumar et al. [56] developed a STEP-compliant framework for the removal of error in manufacturing. It was stated that quality products were consistently produced through the utilization of self-learning algorithm from the previous manufacturing data. In their developed framework, they have proposed knowledge based analyser for analyses of process control data and provides feedback to compensate dynamic and static errors which help to improve surface quality of the components using data mining. Xiao et al. [57] developed a prototype of CAD/CAM/CNC system complaint with STEP-NC based on CATIA Platform having an open structure. For generation of an informative and truthful simulation, Kadir and Xu [58] proposed the High-Fidelity Machining Simulation system, which utilized high-level data of STEP and STEP-NC model for supporting the smart machining environment. They streamed the data acquired from sensor of machine tools and network protocol through MTConnect. Their proposed system provided different modes of machining simulation process, such as pre-machining, machining simulation and post-machining.

Nassehi et al. [59] developed a prototype for the generation of adaptive global manufacturing system for utilization of STEP-NC. The developed prototype used the STEP-NC interoperable process plan to generate NC code. G-M code was created from ShopMill MPF format of feature based information from the STEP-NC code for the prismatic parts. Machining system based on closed loop and object orientation, Hu et al. [60] proposes a system for integrated information flow. For real time machining condition and online inspection, they have also developed an open CNC controller (HITCNC) for interpreting high level information. Before taking decision on machining, its assessment of different process can be done using its environmental analysis so that sustainable manufacturing process would significantly improve [61]. Yosuf et al. [62] discussed the efficient capabilities of STEP-NC than G-M and developed STEP-NC code generator (GENMILL) which focussed on the process of milling based on Example 1 of ISO 14,649 in Part 11 format. Their system was developed on concept of structured methodology and utilized development model of software prototyping. Using JSDAI™, a prototype system for CAD/CAPP/CAM based on Servers (internet) was proposed by Benavente et al. [63] by utilizing different databases of machining conditions and tools. Zhang et al. [64] had implemented STEP-NC program processor and simulator to represent high-level machining simulation system using STEP-NC generator with simulator for STEP-NC. Um et al. [65] implemented the squash algorithm method in the CAD model and performed additive manufacturing of part.

4 Work related to CNC system and manufacturing using STEP-NC (AP238)

Software such as STEP-NC Machine™, ST-Developer™, etc., developed by STEP Tools In[66]. has given opportunity to researchers for generating and testing AP238 codes and also used to convert these STEP-NC code to proprietary code format of various controller. To demonstrate grooving operation Xu [67] (2006) used 6 K motion controller and Compu Cam's motion control system to retrofit the CNC machine. For generating STEP-NC code, STIX (STEP Index Library) was used, which is a library of C++classes related to manufacturing. Lukic et al. [68] describe two scenarios in real manufacturing to implement STEP-NC. STEP Tool Inc. developed STEP-NC programs were applied in first scenario and for customized manufacturing of part STEP-NC programs were concluded in second scenario. Liu et al. [69] used STIX and ST-Developer™ to read STEP-NC file for developing planner and interpreter module for STEP-NC code. However, in their article, it was mentioned that only partial programming using C++was completed and they also represented GUI for part model and tree structure of work plan of AP238 file. Srivastava and Komma [70] developed AP224 based data extractor to generate STEP-NC code in AP238 format.

Different optimized machining models for aerospace part of 'fishhead' were developed by Hardwick [71] using Part 28 such as tolerance model, tool path model, etc. Kubota et al. [72] utilized the concept of STEP-NC for obtaining and analysis of data from knowledge database using machine tool digital twin and developed a framework for data exchange for machining process. Rodrigueza and Alvaresa [73] implemented STEP-NC approach in additive manufacturing and simulated 3D printer model on STEP-NC Machine. Zivanovic et al. [74] developed a G-M code based RoboSTEP-NC module for simulation and actual machining through 5-axis robot. Srivastava and Komma [75] developed a Java interface using JSDAI for integration of CAD/CAM and successfully simulated test parts on STEP-NC Machine. Slavkovic et al. [76] represented different tasks of robot machining through indirect programming using STEP-NC format. They have converted STEP-NC AP238 code to specified robot programming using their developed robot language convertor. Alvares et al. [77] simulated different models of STEP-NC in AP238 and ISO 14,649 on different software and performed real time robotic machining using LinuxCNC. Hardwick et al. [78] described STEP

Part 21 Edition 3, to represent its compatible enhancement to Part 21 and it utilized linking of data and intelligence. This new edition added URI's and JavaScript which was used to enable crowd sourcing of massive product data models. Sample STEP-NC codes given in ISO 14,649 gave the researchers, an opportunity to understand the STEP-NC format. Kramer et al. [79] experimentally proved with the Example 1 of ISO 14649-11 in both ARM and AIM representations that AIM model takes longer time for interpretation and is more complex than ARM model.

5 Work related to industry 4.0, industrial IoT and product data exchange using STEP-NC

On the basis of recommendation of owners of the factories digital twins were created for them using web based form by Lu and Xu [80]. Digital twins were created for three levels of industry i.e. enterprise, factory and equipment using OWL and Jena modelling language to provide flexibility in process virtualization. Some of typical characteristics of intelligent and autonomous Machine Tool 4.0 system were discussed by Xu [81] using integration of different layouts of machine tool and its cyber- production system for generating CNC controller which are intelligent and robust. An initiative should be taken by manufacturing firm to implement either of STEP-NC formats. Hence, SWOT analysis and strategic planning was done by Cha et al. [82] to incorporate organization based on STEP-NC and STEP based standards. Collaborative product exchange mechanism was proposed by Wang and Xu [83] using Distributed Interoperable Manufacturing Platform (DIMP). Data localization/integration mechanism was used by user to access the specific scope of data of STEP and STEP-NC, so that it did not affect rest of the user. Data traceability and confidentiality of different users can be maintained using this representation of data packet and data integrity of STEP. Saif et al. [84] developed a machine vision inspection interpreter system based on STEP-NC and used for inspecting features.

6 Review work related to STEP and STEP-NC

Various researchers also reviewed past work related STEP and STEP-NC. Zhao et al. [85] discussed a typical STEP-compliant manufacturing environment, which was used in integrating the following two systems: (i) first system was used to create native data in getting information needed to machine components on specific machine tool (ii) second system performed optimization of different machining process parameters from the first system. They also reviewed different works based on areas of feature generation, micro and macro process planning and machining execution. Hardwick et al. [86] emphasis importance of AIM format in focussing different structure of models in the ARM format. Saif et al. [87] have given review on different technologies incorporating inspection planning and manufacturing using STEP/STEP-NC and feature based methods. Hardwick and Loffredo [88] described various work related to the testing of STEP-NC AP-238 code for its deployment in the USA. They also showed various test related to STEP-NC AP-238, which allow several CAM systems to send data of 5-axis tool path to various CNC machines without using any post processor. Lu et al. [89] did the critical review on smart manufacturing using authentic vision, concept and their impact on process automation. Xu et al. [90] differentiated AIM and ARM models of STEP-NC and mentioned the possibility of AIM model for full integration of CAD and CAM. Latif et al. [91] have given emphasis to use either ISO 14,649 or ISO 10303-238 format over convention G-M codes in their review article on different technologies of CNC, CAD, CAPP, CAM, STEP, STEP-NC, etc.

Summary of some of the important articles based on STEP and STEP-NC in chronological order is given in Table 1. STEP based data exchange formats benefits to increase effectiveness of Smart Manufacturing. STEP-NC AP238 format utilizes the STEP generic resource (Part 41, Part 42, etc.), which make it one of effective data representation format and can be used throughout of the product life cycle. Due to bi-directional flow of information, compatibility with other application protocols and data preservation STEP-NC AP238 format can effectively use for the scenario of Industry 4.0 or Smart Manufacturing.

7 Conclusions

Smart Manufacturing or Industry 4.0 is transformation of manual labour-mechanization-automation-computer based automation-computer based automation including sensors, servers, internet (or IIoT). Efficiency of the production system and quality of product manufactured are increased using these newer paradigms. It also includes rich set of data to be exchanged through servers between different system and equipment in a proper manner, so that data would not leak to other. This would also include an efficient cyber system, so that exchanged or stored data would not be accessible to hackers. For smart technologies, smart data format would be also required. STEP based methods or formats can be used as sustainable and efficient formats for data exchange between these technologies.

Neutral file representation of product data model according to STEP standard is a promising approach for exchange of product information without losing any information and

Table 1 Summary of some of important review articles based on STEP-NC

Article	Features of the system	Method/language of implementation	Limitations
Suh and Cheon [37]	Intelligent CNC system	PC based CNC using software interfaced with hardware and interface board	Only conceptual framework presented
Suh et al. [32]	Developed shop floor programming known as Pos-SFP for recognition of features, process plans and generate and verify STEP-NC codes	C++, OpenGL	Intelligence, optimization, interactive feature recognition
Newman et al. [43]	STEP-compliant Agent based Manufacture system (AB-CAM)	IDEF0 representation, Java	Determination of optimal toolpath trajectories
You and Lin [15]	Java based CAPP system	Enterprise Java Beans (EJBs)	Setup planning
Medani and Ratchev [21]	Agent-based manufacturability environment based on STEP AP224	CORBA, XML	Setup planning, toolpath generation
Xu [67]	Retrofitted the existing CNC with Compu Cam's motion control system	C++, STIX	STEP to NC controller
Shin et al. [31]	G2STEP system for automatic conversion to STEP-NC from G-code.	C++, OpenGL	Only for turned part
Nassehi et al. [41]	Integrated Platform for Process Planning and Control (IP3AC) to support CAPP and CAM using STEP Data Access Interface	Java, SHOPMILL	Uses non STEP intermediate file
Yang and Xu [44]	STEP-NC compliant machine tool data model	EMCO Concept Milling machine, XML	Platform based on internet
Zhao et al. [47]	Closed loop machining process to integrate CAD, process planning, machining and inspection chain	Express Engine 3.2 software	Inspection process planning and inspection data analysis
Hardwick [71]	Third generation of STEP to aggregate information	C++	Use of non STEP intermediate format
Yosuf et al. [62]	STEP-NC code generator (GENMILL)	Visual Basic	Only for Example 1 of ISO 14,649
Amaitik [22]	Intelligent CAPP system, fuzzy logic, neural network	Integrator environment with AutoCAD, XML	Toolpath generation
Zhang et al. [64]	High-level machining simulation system	Different prototypes using IDEF0 modelling	Modularized process planning, machining simulation and execution
Benavente et al. [63]	CAD/CAPP/CAM prototype system	JSDAI™	Confined to 3 axis machining, Used ARM format
Hardwick et al. [86]	Description of STEP Part 21 Edition 3	URI's and JavaScript	Demonstration to community. Use of non STEP intermediate format
Xu et al. [53]	Canonical machining functions to convert G-code to STEP-NC	C++	Detection of turning or milling type and number of axis in the program. Use of non STEP intermediate format
Zhang et al. [55]	Hybrid feature recognition from the G-M codes and represented it to STEP-NC.	Simulation	Shop floor information feedback, knowledge- based CAPP systems. Use of non STEP intermediate format
Kumar et al. [23]	STEP XML based feature extractor, interactive feature recognition	Autodesk Inventor environment, XML	Selection, fixture selection, setup selection and automatic generation of NC part
Shin et al. [42]	Virtual machining model (STEP2M i.e. STEP to MTconnect)	Java, Tomcat	Upto 2-axis turning machining
Yusof and Latif [45]	STEP-NC and G-M codes to Open Architecture Control (OAC) based CNC machine	LabVIEW	GUIs development, simulation, toolpath generation
Hu et al. [60]	Closed loop machining system (CLMS)	C++	Real-time and online machining process control algorithm
Xu [82]	Machine Tool 4.0	Servers, Internet of Things (IoT)	Require a collective effort from the machine tool manufacturers, users and researchers
Rodrigueza and Alvaresa [73]	3D printer model on STEP-NC Machine	C#	STEP-NC machining information is not available
Kubota et al. [72]	Data exchange using machine tool digital twin	Server, OPC Unified Architecture	System is not autonomous
Saif et al. [79]	Machine vision inspection interpreter system	Cyber, IoT	System is not autonomous

for seamless integration of different sections of a manufacturing organization or different organizations. Human readable and computer interpretable structure of STEP based models make the information exchange faster and error free. Different application areas of STEP in the form of STEP APs provide wide platform for their applicability over different sections of manufacturing domain. STEP AP203 (or its recent version AP242) consist of 2D or 3D part information of a product in a systematic and object oriented approach. Feature description of STEP in AP224 format presents a well-defined structure that can be used for selection of feature based CAPP parameters. CAD model data and manufacturing features information shall be used to generate code for computer control of machine tools. STEP-NC evolved as a superior format for computer control of machine tools as compared to the conventional G-M codes. G-M code provides only the relative movement of tool with respect to work piece. STEP-NC code contains not only the tool movement information but also information of operations, geometry of work piece, tool, manufacturing features etc. Information flow is in bi-direction way in STEP-NC or we can say that, the efficient feedback is maintained from equipment and controller through computer in STEP-NC. Therefore, integration of CAD/CAPP/CAM is seamlessly and efficiently provided through STEP-NC. There are two formats of STEP-NC: ISO 14,649 (ARM format) and ISO 10303-238 (AIM format). When compared to ARM format, AIM format of STEP-NC has rich set of features and is more sustainable, which can be used throughout the life cycle of product, as it uses STEP generic resources for building a particular model.

It can be concluded from the literature review that STEP-NC AP238 seems to be a promising format to be used for data exchange in Industry 4.0 or Smart Manufacturing. Majority of researcher used this format for development of their manufacturing system involving the concept of internet, servers, sensors, IIoT, etc. Standard STEP data access interface using programming language of C + and Java were created by majority of researcher. AP238 also support data representation in XML format also, which can be used in developing web based technologies using server. AP238 support interface development in C + or Java either in clear text encoding (Part 21) and XML encoding (Part 28). It also used data description in terms of STEP integrated generic resources, which helps in efficient data representation of product model.

The concept of Industry 4.0 can increase the productivity of the industry. The time consuming in manufacturing would be decreased using data exchange through servers or internet, which can also increase overall efficiency of the system. This technology also implies the concept of artificial intelligence and machine learning which would create the system or set of equipment as autonomous, can take their own decisions and rectify errors, judge production requirements according to market fluctuations. Role of sensors for sending feedback to the integrated system through internet are very crucial and important for development of smart manufacturing system. The microcontroller and microprocessor connected to sensors are based on IDE, i.e. data input/output system, which can work on different programming language and can be linked to database available on server through internet. Overall the scenario of Industry 4.0 or smart manufacturing can be best utilized through AP238 format of STEP-NC.

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